Shapes & Symmetry

Detailed Lesson Plan — Mathematics

Subject	Mathematics — Geometry
Duration	5 Sessions × 45 minutes (225 minutes total)
Curriculum	2D Shapes, Properties of Shapes, Lines of Symmetry
Resources	Slides deck · Printed worksheets · Manipulatives · Ruler

Unit Description

This five-session unit introduces Grade 2 students to the world of two-dimensional shapes and the concept of symmetry. Using the colourful *Shapes & Symmetry — Let's Explore!* slide deck as a visual anchor, learners will name and describe circles, squares, rectangles, triangles, and semicircles; count sides and corners; locate examples in their environment; and identify lines of symmetry in both shapes and real-world objects such as butterflies. Activities range from whole-class discussion and interactive games to partner work, art integration, and printable worksheets, ensuring all learning styles are reached.

Learning Objectives & Curriculum Standards

By the End of This Unit, Students Will Be Able To:

Knowledge

- Identify and name the five 2D shapes: circle, square, rectangle, triangle, and semicircle.
- Define the terms *sides* (straight or curved lines forming a shape) and *corners* (points where two sides meet).
- State the number of sides and corners for each shape studied.
- Explain symmetry using the 'fold-in-half' concept.

Skills

- Count sides and corners on any given 2D shape.
- Draw each of the five shapes freehand.
- Draw at least one line of symmetry on a shape.
- Locate shapes in the classroom and school environment.
- Match real-world objects to their corresponding shape.

Application

- Create a 'shape picture' using multiple shapes.
- Sort objects by shape category.
- Complete worksheets requiring shape identification and symmetry.
- Distinguish between symmetrical and non-symmetrical shapes.

Curriculum Alignment

CCSS.MATH.CONTENT.2.G.A.1	Recognize and draw shapes having specified attributes (number of angles, faces, sides).
CCSS.MATH.CONTENT.1.G.A.3	Partition circles and rectangles into two and four equal shares; understand symmetry.
NCERT Class 2 — Chapter 6	Shapes around us — identification, naming, and properties of common 2D shapes.
State Standard G-2.2	Students identify, sort, and classify shapes by attributes including sides and corners.

Materials, Resources & Key Vocabulary

Materials Required

Teacher Materials	Student Materials	Classroom Setup
Shapes & Symmetry slide deck (9 slides, printed / projected)	Pencils, crayons, coloured markers	Projector or interactive whiteboard
Printed shape flashcards (A5, laminated)	Scissors, glue sticks	Enough floor space for circle time
Ruler / metre stick for demo	Printed worksheets (1 per student per session)	Small mirrors (1 per pair) for symmetry activity
A large butterfly cut-out (for symmetry demo)	Rulers or straight-edge cards	Shape sorting trays or hoops
Mini whiteboards × class set	Mini whiteboards + markers	Display board for shape chart

Key Vocabulary

Shape	A flat figure made of lines or curves.
Side	A straight or curved line that forms part of the boundary of a shape.
Corner	The point where two sides of a shape meet (also called a vertex).
Circle	A perfectly round shape with no corners and no straight sides.
Square	A shape with 4 equal sides and 4 corners.
Rectangle	A shape with 4 sides (2 long, 2 short) and 4 corners.
Triangle	A shape with 3 sides and 3 corners.
Semicircle	Half of a circle; has 1 curved side and 1 straight side.
Symmetry	When a shape can be folded so both halves match exactly.
Line of Symmetry	An imaginary line that divides a shape into two mirror-image halves.

Session 1 · What Are Shapes? — Introduction (Slides 1–2)

Objective	Introduce shapes concept; activate prior knowledge; learn the terms 'sides' and ' corners'.
Differentiation	Support: labelled shape mat on desk. Extension: students list 2 real-world objects for each shape.

Phase	Description
Warm-Up (5 min)	Begin with a 'Shape Hunt': ask students to look around the classroom and call out any shape they can see. Record responses on the board. Ask: 'What makes a square different from a circle?'
Direct Instruction (10 min)	Display Slide 1 (title). Introduce the big idea: shapes are everywhere, they have sides and corners. Show Slide 2. Read each bullet with the class. Point to real objects (window = rectangle, clock face = circle).
Guided Practice (15 min)	Flashcard flash: teacher holds up a shape card; students write the name on mini-whiteboards and hold up on count of 3. Reinforce correct answers with the slide imagery.
Independent Activity (10 min)	Worksheet 1: 'Name That Shape!' — students write the name beneath 8 shape outlines and colour each one a given colour (circle = red, square = blue, triangle = yellow, rectangle = green).
Wrap-Up (5 min)	Exit ticket: students draw one shape and write how many sides and corners it has. Share two or three with the class.

Session 2 · Meet the Shapes — Properties Deep Dive (Slides 3–4)

Objective	Describe each shape by number of sides and corners; introduce semicircle.
Differentiation	Support: provide pre-filled chart template. Extension: explore hexagon and pentagon as bonus shapes.

Phase	Description
Warm-Up (5 min)	Quick quiz — teacher says a clue ('I have 3 sides and 3 corners, who am I?') and students respond on whiteboards. Do 6 rounds.
Direct Instruction (12 min)	Display Slide 3 (Meet the Shapes). Discuss each of the four 2D shapes shown. Move to Slide 4: introduce semicircle — 'half a circle'. Discuss how many curved sides (1) and how many straight sides (1) it has.
Guided Practice (13 min)	Class completes a 'Shapes Properties Chart' together on the board: Shape | Sides | Corners | Example in Real Life. Students copy it into their maths journals. Prompt for real-life examples from the slide (cookie = circle, sandwich = square).
Partner Activity (10 min)	Pairs receive a bag of plastic/foam shapes. One partner picks a shape without showing it and gives clue sentences ('My shape has 4 equal sides…'). Partner guesses. Swap roles every 2 minutes.
Wrap-Up (5 min)	Review the chart together. Ask: 'Which shape has the most corners? Which has none?'

Session 3 · Shapes Around Us — Real-World Connections (Slide 5)

Objective	Connect 2D shapes to everyday objects; become 'shape detectives'.
Differentiation	Support: colour-coded shape reference card. Extension: students count and record total shapes used in their picture.

Phase	Description
Warm-Up (5 min)	'Shape of the Day' — teacher holds up a photograph and students name the dominant shape. Use images from around the school.
Direct Instruction (8 min)	Display Slide 5 (Shapes Around Us / Circles Everywhere). Read the slide. Discuss the example objects: cookie = circle, ball = circle. Ask students to extend the list.
Class Walk (15 min)	Classroom / corridor shape walk. Each student carries a 'Shape Detective' clipboard sheet divided into 5 columns (one per shape). Students tally each shape they spot. Return to class and compare findings.
Art Activity (12 min)	Students create a 'Shape Town' picture using cut-out shapes (pre-cut from coloured paper). Must use all 5 shapes. Label each shape used.
Wrap-Up (5 min)	Gallery walk of 'Shape Town' pictures. Students leave a sticky-note comment on a partner's work naming one shape they can see.

Session 4 · What Is Symmetry? — Mirror Magic (Slide 6)

Objective	Define symmetry; identify symmetrical objects in nature; use a mirror to explore symmetry.
Differentiation	Support: dotted grid paper. Extension: students create their own symmetrical ink-blot art (fold paper, add paint, unfold).

Phase	Description
Warm-Up (5 min)	Show a large butterfly cut-out. Ask: 'What do you notice about the wings?' Fold the butterfly in half to demonstrate matching sides.
Direct Instruction (12 min)	Display Slide 6 (What Is Symmetry?). Introduce the vocabulary: symmetry, line of symmetry, mirror image. Use the butterfly image on the slide as the main example. Demonstrate with a mirror: place mirror along the fold line of a picture.
Guided Practice (10 min)	Distribute small mirrors. Teacher displays simple shapes on the whiteboard. Students place their mirror at different positions and tell a partner whether both sides match. Discuss: does a circle always have symmetry? Does a random squiggly line?
Independent Activity (13 min)	Worksheet 2: 'Mirror Magic' — students draw the missing half of 6 symmetrical pictures (butterfly, heart, house, leaf, face, star). Requires a ruler to find the fold line.
Wrap-Up (5 min)	Students share one thing they found surprising about symmetry. Teacher records statements on a 'What We Learned' anchor chart.

Session 5 · Symmetry in Shapes — Lines of Symmetry & Review (Slides 7–9)

Objective	Draw lines of symmetry on geometric shapes; consolidate all unit learning.
Differentiation	Support: word bank on worksheet. Extension: students draw a shape with more than 2 lines of symmetry and explain their thinking.

Phase	Description
Warm-Up (5 min)	Rapid-fire review: teacher shows shape flashcards, students call out name, sides, corners as fast as possible.
Direct Instruction (10 min)	Display Slide 7 (Symmetry in Shapes). Show lines of symmetry on circle, square, and rectangle. Ask: 'How many lines of symmetry does a square have? A circle?' Introduce the concept that some shapes (like squares) have more than one line of symmetry.
Guided Practice (10 min)	Using Slide 8 (Let's Practice), work through each bullet as a class: find shapes in the classroom, draw a line of symmetry on a whiteboard shape, match shapes to names, count sides and corners on projected shapes.
Assessment Activity (15 min)	Worksheet 3 (Unit Review): (a) Name and colour 6 shapes. (b) Draw the line of symmetry on 4 shapes. (c) Write how many sides & corners each shape has. (d) Circle the symmetrical objects from a set of 8 pictures.
Wrap-Up / Celebration (5 min)	Display Slide 9 (Great Job, Shape Explorers!). Class celebration: students call out one shape fact they remember. Teacher highlights key vocabulary on the anchor chart.

Assessment, Rubric & Teacher Notes

Formative Assessment (Ongoing)

- Exit tickets at the end of Sessions 1 and 3 (draw a shape; name the shape).
- Observation during group and partner activities — anecdotal notes on shape identification.
- Mini-whiteboard responses during rapid-fire reviews.
- Worksheet 1 (Session 1) and Worksheet 2 (Session 4) — collected and reviewed.

Summative Assessment (Session 5 — Worksheet 3)

Criterion	Exceeds (4)	Meets (3)	Approaching (2)	Beginning (1)
Shape ID	Names all 5 shapes confidently + gives a real-world example.	Names all 5 shapes correctly.	Names 3–4 shapes correctly.	Names fewer than 3 shapes.
Properties	Accurately counts sides & corners for all shapes + explains.	Correctly states sides & corners for all shapes.	Correct for 3–4 shapes.	Correct for fewer than 3 shapes.
Symmetry	Draws multiple lines of symmetry; explains the concept clearly.	Draws the correct line of symmetry on all 4 shapes.	Draws symmetry correctly on 2–3 shapes.	Attempts but places line incorrectly.
Real-World	Identifies all 8 pictures correctly; explains choice.	Identifies 6–8 correctly.	Identifies 4–5 correctly.	Identifies fewer than 4.

Teacher Notes & Pedagogical Tips

- **Prior Knowledge:** Check that students can count to at least 8 and understand 'half'. Use informal pre-assessment through the Session 1 warm-up shape hunt.
- **Manipulatives First:** Whenever possible, let students hold and trace physical shape blocks before working abstractly on paper. Kinaesthetic experience deepens conceptual understanding.
- **Symmetry Misconception:** Many students initially believe any line through a shape creates symmetry. Counter this explicitly by showing a diagonal cut through a rectangle that does *not* produce two equal halves.
- **Language Support (EAL/D):** Post a bilingual vocabulary chart. Pair students strategically for partner activities. Allow drawing as an alternative to written explanations.
- **Cross-Curricular Links:** Art (Session 3 shape pictures); Science (symmetry in nature — leaves, flowers, insects); English (shape poems, descriptive writing).
- **Home Connection:** Send home a 'Shape Spotter' family activity sheet after Session 3. Families photograph shapes at home; students share findings at the start of Session 4.
- **Display Ideas:** Create a classroom 'Shape Wall' with labelled examples. Add student-drawn shapes and real-world photographs throughout the unit.

Slide Deck Reference

Slide 1	Title — Shapes & Symmetry: Let's Explore!	Session 1 opening

Slide 2	What Are Shapes? (sides, corners, classroom hunt)	Session 1
Slide 3	Meet the Shapes! (circle, rectangle, triangle)	Session 2
Slide 4	More Shape Friends! (semicircle, sides, corners)	Session 2
Slide 5	Shapes Around Us / Circles Everywhere	Session 3
Slide 6	What Is Symmetry? (butterfly example)	Session 4
Slide 7	Symmetry in Shapes (line of symmetry definition)	Session 5
Slide 8	Let's Practice! (review activities)	Session 5
Slide 9	Great Job, Shape Explorers! (celebration)	Session 5

End of Lesson Plan · Shapes & Symmetry